ADITYA PRASANNA MOGARE

Los Angeles, CA 90007, United States | mogare@usc.edu | +1(213)756-4828 | Linkedin |Portfolio | GitHub | Publication

EDUCATION

University of Southern California, Los Angeles **August 2024 - May 2026**
Master of Science in Computer Science **GPA 3.4/4.0**
Courses: Deep Learning, Natural Language Processing, Analysis of Algorithms
University of Mumbai, India **June 2020 - June 2024**
Bachelor of Engineering in Computer Engineering **GPA 9.63/10**
Courses: Algorithms & Data structures, Computer networks, Operating Systems, Database Management

SKILLS

- **Programming Languages:** Python, C++, C, JavaScript (ES6+), Java, SQL, HTML, CSS
- **Full-Stack & Frameworks:** React.js, Angular, Node.js, Express.js, Flask, A-Frame
- **Machine Learning & AI:** LLM fine-tuning (LoRA), Transformers, CNNs (VGG, ResNet, Inception), Gradient Boosting, RAG Pipeline, Semantic Search
- **Simulation & System Design:** High-throughput web systems, Real-time collaboration SDKs, Distributed systems, Performance profiling
- **DevOps & Tools:** Docker, Kubernetes, Jenkins, Git (GitHub, GitLab), VS Code, AWS (EC2, S3, Lambda)
- **Databases:** PostgreSQL, MongoDB, MySQL, SQLite
- **Software Engineering Practices:** CI/CD, Agile & DevOps workflows, Version Control, Test Automation

PROFESSIONAL EXPERIENCE

INTERDEPENDENT, Onsite | Software Engineering Intern **May 2025 – August 2025**

- Developed **scalable full-stack solutions** with **React.js**, **Python (Flask)**, and **RESTful APIs**, enabling real-time synchronization of creative planning workflows via Miro SDK
- Built **distributed system components** leveraging event-driven architecture and webhooks for simulation-like environments
- Fine-tuned **Transformer-based LLMs** for **automated screenplay analysis**, integrating structured data inputs for **budgeting/scheduling emulation** — simulating creative pipelines analogous to flight systems
- Contributed to performance profiling and deployment using **Docker** and **Kubernetes**

Research Assistant | USC Viterbi School of Engineering **Feb 2025 – June 2025**
Prof. Anita Penkova | Project: Diabetic Retinopathy

- Designed **multi-class classification pipelines** with **high-fidelity medical image simulations** (OCTA) using **CNNs (VGG, ResNet)**.
- Achieved a **+5% AUC increase** over baseline by optimizing **image preprocessing** (noise reduction, normalization, and augmentation).
- Deployed models on **AWS EC2** using **Docker**, demonstrating familiarity with high-performance computing workflows.

HiringTek, Remote | Software Engineering Intern – AI & Full Stack Development **Aug 2022 - Aug 2023**

- Developed a GPT-powered **question generation engine** with Flask and MongoDB, increasing content throughput **10×**
- Automated Kubernetes deployment across 4 environments, reducing setup and test cycles by **25%**.
- **Contributed to the Test Automation** pipeline by managing 30+ QA scripts and ensuring 95 %+ test coverage for mission-critical modules like QA Bank, Evaluation, and Candidate workflows.

PROJECTS

LurisQA (Python, Transformers, LoRA, SentenceTransformers, Faiss, RAG) (Github Link)

- Deployed a scalable **legal domain question-answering (QA) system** by fine-tuning state-of-the-art open-source LLMs (e.g., LLaMA, Mistral) using Low-Rank Adaptation (LoRA), achieving a 4× reduction in GPU memory footprint and enabling cost-effective model training and deployment.
- Engineered end-to-end ML pipelines to ingest, preprocess, and index 100K+ U.S. legal documents, implementing semantic search vector stores and SentenceTransformers embeddings, resulting in sub-second document retrieval performance

NotesMadeEasy (Python, Streamlit, LangChain, Ollama, LiteLLM, Markdown, OpenCV) (Github Link)

- Built and deployed an AI-powered note-taking assistant that extracts structured notes from lectures, PDFs, and YouTube videos using LLMs and LangChain pipelines.
- Integrated local LLMs (Gemma, LLaMA3) via Ollama and LiteLLM for fast, offline access with semantic chunking and real-time PDF parsing.
- Designed UI with Streamlit, added Pomodoro timer, Markdown export, and citation features to enhance productivity.

EcomForecast: Empowering E-Commerce through Advanced Analytics and Forecasting (Github Link)

- **Developed EcomForecas**t, a platform using advanced time series models (SMA, EMA, WMA, LSTM, GRU) for precise sales forecasting.
- Qualified for the final round of presentations for International Confluence on Startups and Innovation (ICCSI-2023 CREST IITM) at IIT Madras Campus (December 13-15, 2023)

RESEARCH PUBLICATION

- Wiley (link): Building a Virtual Reality-Based Framework for the Education of Autistic Kids